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IMITATION COSTS AND PATENTS: AN EMPIRICAL STUDY*

Edwin Mansfield, Mark Schwartz and Samuel Wagner

It has long been recognised that the costs of imitating new products have an important effect on the incentives for innovation in a market economy.¹ As Arrow (1962) and others have pointed out, if firms can imitate an innovation at a cost that is substantially below the cost to the innovator of developing the innovation, there may be little or no incentive for the innovator to carry out the innovation. In their discussions of the innovation process, economists frequently have called attention to the major role played by the costs of imitation, but there has been little or no attempt to measure these costs, to test various hypotheses concerning the factors influencing them, or to estimate their effects. In this paper, we report some findings of what seems to be the first study of this topic. In addition, we present data regarding the effects of patents on imitation costs and on the rate of innovation.

I. IMITATION COSTS AND TIMES

Many economists, including Scherer (1977) and Mansfield *et al.* (1977*b*), have cited the need for empirical studies of imitation costs and imitation times. To carry out such a study, we obtained data from firms in the chemical, drug, electronics, and machinery industries concerning the cost and time of imitating (legally) 48 product innovations. By imitation cost we mean all costs of developing and introducing the imitative product, including applied research, product specification, pilot plant or prototype construction, investment in plant and equipment, and manufacturing and marketing startup.² (If there was a patent on the innovation, the cost of inventing around it is included.) By imitation time we mean the length of time elapsing from the beginning of the imitator's applied research (if there was any) on the imitative product to the date of its commercial introduction.³

The firms were chosen more or less at random from among the major firms in these four industries in the Northeast of the United States, and the new pro-

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¹ For example, see Arrow (1962) and Freeman (1974).

² For definitions of these terms, see Mansfield *et al.* (1971) and Mansfield (1980).

³ For many products, the estimate of imitation cost (and of imitation time) was an average of two estimates provided by the firms, each reflecting a somewhat different set of assumptions.

Table 1
*Imitation Cost (Divided by Innovation Cost) of 48 New Products, by
 Industry and Cost of Innovation*

Imitation cost (divided by innovation cost)	Innovations costing more than \$1 million			Innovations costing less than \$1 million		
	Chemicals	Drugs	Electronics and machinery	Chemicals	Drugs	Electronics and machinery
(A) Number of new products						
Less than 0.20	1	1	1	1	0	0
0.20 and under 0.40	0	3	0	0	0	2
0.40 and under 0.60	1	1	2	5	0	0
0.60 and under 0.80	2	5	0	2	0	4
0.80 and under 1.00	2	3	1	2	1	1
1.00 and over	2	2	1	1	0	1
Total	8	15	5	11	1	8
(B) New products weighted by innovation cost* (%)						
Less than 0.20	3	3	17	15	0	0
0.20 and under 0.40	0	11	0	0	0	34
0.40 and under 0.60	†	1	53	46	0	0
0.60 and under 0.80	44	54	0	13	0	36
0.80 and under 1.00	15	21	9	22	100	18
1.00 and over	38	9	22	4	0	11
Total†	100	100	100	100	100	100

* The weighted number of new products is expressed as a percentage of the column total.

† Because of rounding errors, items may not sum to column total.

‡ Less than 0.5.

ducts were chosen more or less at random from among those introduced recently by these firms.¹ In 34 of the cases, the new product had already been imitated, so the data are based on actual experience. In the remaining 14 cases, no imitator had appeared as yet, but the innovating firm provided us with detailed estimates that were regarded as being reliable.² Also, in all 48 cases,

¹ We say that the firms and innovations were chosen 'more or less' at random because a table of random numbers really was not used. In some cases, the sample was a systematic sample from a list; in other cases, there were so few innovations in the time frame we selected that the choice was automatic. Both successful and unsuccessful new products were included in the sample. About half of the innovations turned out to be successful (in the sense that they were relatively profitable to the innovator). A high percentage of the firms we approached agreed to co-operate so there was little problem of nonresponse. A considerable number of in-depth interviews were held with major officials of each firm to insure that the data were as accurate as possible. Nineteen of the innovations occurred in the chemical industry (including petroleum refining); sixteen occurred in the drug industry; ten occurred in the electronics industry; and three occurred in the machinery industry. Five of the new products were first introduced before 1960; fifteen were first introduced during the 1960s; and 28 were first introduced during 1970-6.

² In each such case, the innovating firm provided an estimate of how much money and time it would have cost the most likely (and most efficient) imitator to have imitated the product. In some cases, it was assumed that the time-cost combination that would have been chosen by the imitator was midway between the least-cost combination and the least-time combination. According to the firms, these estimates are unlikely to be very wide of the mark. If we had dropped these cases from the sample, we would have risked the introduction of a serious bias. Thus, this seemed to be the best feasible procedure. This procedure also was used in other cases, and seemed accurate enough for present purposes.

Table 2
*Imitation Time (Divided by Innovation Time) of 48 New Products, by
 Industry and Cost of Innovation*

Imitation time (divided by innovation time)	Innovations costing more than \$1 million			Innovations costing less than \$1 million		
	Chemicals	Drugs	Electronics and machinery	Chemicals	Drugs	Electronics and machinery
(A) Number of new products						
Less than 0.30	1	2	1	2	0	0
0.30 and under 0.50	1	5	1	2	0	2
0.50 and under 0.70	1	3	1	4	1	2
0.70 and under 0.90	3	0	0	1	0	1
0.90 and under 1.10	1	3	1	1	0	1
1.10 and over	1	2	1	1	0	2
Total	8	15	5	11	1	8
(B) New products weighted by innovation cost* (%)						
Less than 0.30	3	10	35	16	0	0
0.30 and under 0.50	2	28	17	36	0	16
0.50 and under 0.70	35	24	19	27	100	23
0.70 and under 0.90	22	0	0	16	0	14
0.90 and under 1.10	18	16	22	4	0	18
1.10 and over	19	23	9	2	0	30
Total†	100	100	100	100	100	100

* The weighted number of new products is expressed as a percentage of the column total.

† Because of rounding errors, items may not sum to column total.

data were obtained from the innovating firm concerning the costs of the innovation, as well as the time it took to bring the innovation to market (from the beginning of applied research to the date of its commercial introduction).

The innovations included in the sample vary widely in importance, but practically all are major new products that are central to the innovators' activities, not peripheral to their main business. For 30 of the 48 products, the innovation cost exceeded \$1 million; for 12 products, it exceeded \$5 million. About 70% of the innovations were patented. In only one case did the innovator license the imitating firm. In all other cases, the imitator received no help from the innovator. In general, these firms did not appear to be very interested in licensing these innovations, at least in the relevant time period, often because they did not feel it was in their interest to encourage potential rivals.

On the average, the ratio of the imitation cost to the innovation cost was about 0.65, and the ratio of the imitation time to the innovation time was about 0.70. As shown in Tables 1 and 2, there is considerable variation about these averages. In about half of the cases, the ratio of imitation cost to innovation cost was either less than 0.40 or more than 0.90. In about half of the cases, the ratio of imitation time to innovation time was either less than 0.40 or

more than 1.00. Products with a relatively high (low) ratio of imitation cost to innovation cost tended to have a relatively high (low) ratio of imitation time to innovation time.¹

It may come as a surprise that imitation cost was no smaller than innovation cost in about one seventh of the cases. This was not due to any superiority of the imitative product over the innovation. Instead, in a substantial percentage of these cases, it was due to the innovator's having a technological edge over its rivals in the relevant field. Often this edge was due to superior 'know-how' – that is, better and more extensive technical information based on highly specialised experience with the development and production of related products and processes. Such know-how is not divulged in patents and is relatively inaccessible (at least for a period of time) to potential imitators.

Based on these results, it appears that innovators routinely introduce new products despite the fact that other firms can imitate these products at about two thirds (often less) of the cost and time expended by the innovator.² In some cases, this is because, although other firms could imitate these products in this way, there are other barriers to entry (for example, lack of a well-known brand name) that discourage potential imitators. But to a greater extent (at least in this sample), it seems to be due to a feeling on the part of the innovators that, even if imitators do begin to appear in a relatively few years, the innovation still will be profitable.

II. THE IMITATOR'S TIME-COST TRADEOFF FUNCTION

Clearly, the time it takes a firm to imitate a new product can generally be reduced by spending more money. Each product's imitator was confronted by a time-cost tradeoff function, which is the relationship between the amount spent by the imitator and the length of time it would take to imitate this new product.³ The time-cost combination chosen by the imitator is one point on this tradeoff function. In most cases, we obtained an estimate of the minimum time that the imitator could have taken to carry out the project, and the corresponding cost, as well as an estimate of the minimum cost and the corresponding time. Together with other information, this enabled us to determine the approximate position of the imitator's time-cost tradeoff function for most products. If imitation cost is measured as a percentage of innovation cost and if imitation time is measured as a percentage of innovation time, all of these time-cost tradeoff functions can be plotted on the same graph. When this is done, we find that a product's ratio of imitation cost to innovation cost is a good indicator of how high and how far to the right the product's time-cost tradeoff function is located.

¹ The correlation coefficient between a product's ratio of imitation cost to innovation cost and its ratio of imitation time to innovation time is about 0.8.

² The available data pertain only to innovations that were introduced, not to those where the incentives were insufficient for their introduction. Although they cannot measure the effects of imitation cost on how many innovations were introduced, they provide valuable evidence concerning the characteristics of those that were introduced.

³ For further discussion of time-cost tradeoff functions, see Scherer (1967, 1980), Mansfield *et al.* (1971, forthcoming), and Baldwin and Childs (1969).

Using the time-cost tradeoff function for each product, we computed the arc elasticity of cost with respect to time. The median value of this elasticity is about 0.7, which means that a 1 % reduction in time results in about a 0.7 increase in cost, on the average.¹ These elasticities pertain to the imitation time that is midway between the minimum time and the time corresponding to minimum cost. Since the time-cost tradeoff function is convex, this elasticity increases as imitation time falls (and approaches its minimum value), and decreases as imitation time rises (and approaches the time corresponding to minimum cost). For most products, the minimum-time imitation cost is less than 50 % greater than the minimum imitation cost, but in some cases the difference is 100 % or more. Ordinarily, the minimum-cost imitation time is less than 75 % greater than the minimum imitation time, but in over one-fourth of the cases the difference is 100 % or more (Table 3).

Table 3

Ratio of Minimum-Time Imitation Cost to Minimum Imitation Cost, and Ratio of Minimum-Cost Imitation Time to Minimum Imitation Time, Percentage Distribution of New Products in the Sample

Value of ratio	Minimum-time imitation cost ÷ minimum imitation cost	Minimum-cost imitation time ÷ minimum imitation time
	(% of new products)	
1.00 and under 1.25	36	18
1.25 and under 1.50	33	24
1.50 and under 1.75	6	24
1.75 and under 2.00	6	3
2.00 and under 2.25	12	15
2.25 and under 2.50	6	6
2.50 and over	0	9
Total	100*	100*

* Because of rounding errors, the percentages do not sum to the column total.

III. DETERMINANTS OF IMITATION COSTS

The following hypotheses help to explain the substantial variation among products in the ratio of imitation cost to innovation cost.² First, we would

¹ Estimates of the elasticity of cost with respect to time could be obtained for 39 products. In 5 cases, it was less than 0.25; in 10 cases, it was 0.25 and under 0.50; in 12 cases, it was 0.50 and under 1.00; in 10 cases, it was 1.00 and under 2.00; and in 2 cases, it was 2.00 and over.

² These hypotheses also help to explain the variation among products in the ratio of imitation time to innovation time, since (as pointed out in note 1 on p. 910) a product's ratio of imitation time to innovation time is positively correlated with its ratio of imitation to innovation cost.

For some purposes, one might want to explain the variation among products in the ratio of imitation cost to innovation cost when the ratio of imitation time to innovation time is held constant (at its mean value). Based on our estimates of the time-cost tradeoff functions, we were able to make rough estimates of most products' ratio of imitation cost to innovation cost under these circumstances. When these estimates are used (in place of C_i) in equation (2), the results are relatively unchanged. Thus, the hypotheses discussed in this section can explain the variation among products in both the unadjusted ratio of imitation cost to innovation cost and the ratio of imitation cost to innovation cost when the ratio of imitation time to innovation time is held constant.

expect the i th product's ratio of imitation cost to innovation cost (denoted by C_i) to be inversely related to the proportion of the i th product's innovation cost that goes for research (rather than product specification, pilot plant or prototype, plant and equipment, or manufacturing or marketing startup). An imitator frequently can spend much less time and money on research than the innovator because the product's existence and characteristics provide the imitator with a great deal of information that the innovator had to obtain through its own research. On the other hand, an imitator often has to go through many of the same steps as the innovator with respect to pilot plant or prototype construction, investment in plant and equipment, and manufacturing and marketing startup.

Second, we would expect C_i to be relatively large if the i th product is a new drug where the Food and Drug Administration requires that an imitative product must be tested in much the same way as the innovative product with which it competes. As Kitch (1973) and others have pointed out, this requirement increases the money and time that an imitator must spend. (How much it increases them is a matter of some debate; results on this score are given below.) Third, we would expect C_i to be relatively small if the i th product consists of a new use for an existing material and if some firm other than the innovator has patents on this material. In such cases, the patent holder often can imitate the innovation relatively cheaply and quickly.¹

To test these hypotheses, we assume that

$$C_i = \alpha_0 + \alpha_1 R_i + \alpha_2 D_i + \alpha_3 G_i + z_i, \quad (1)$$

where R_i is the percentage of the innovation cost that went for applied research in the case of the i th product, D_i is a dummy variable that equals 1 if the i th innovation was an ethical drug (subject to regulation of the sort described above) and 0 otherwise, G_i is a dummy variable that equals 1 if the i th product was a new use for an existing material on which a firm other than the innovator holds patents and 0 otherwise, and z_i is a random error term. Data were collected regarding R_i , D_i , and G_i for 28 of the products in our sample,² and we computed least-squares estimates of the α 's. The results are

$$C_i = 0.838 - 0.00684R_i + 0.310D_i - 0.536G_i. \quad (\bar{R}^2 = 0.49) \quad (2)$$

(11.8) (3.80) (2.31) (2.41)

The t -ratios are given in parentheses.

¹ In one case in our sample, an agricultural chemical firm found a new use for a particular chemical. This firm's supplier held a patent on the broad group of chemicals of which this chemical was part, but was unaware of this new use. When the innovator marketed the chemical for this new use, the supplier could (and did) imitate the innovation quickly and at a low cost.

Another factor we would like to include in equation (1) is the extent of the imitator's technological capabilities and know-how, relative to those of the innovator, but we could find no way to measure this factor adequately.

² The sample was composed of two subsamples, each of which was collected quite independently from the other. In the subsample for which Schwartz was particularly responsible, we obtained the data needed to test these hypotheses. In the other sub-sample, for which Wagner was particularly responsible, we did not obtain these data, but focused attention on other questions instead. Equation (4) is also based only on the former subsample. One product was not included in equation (2) because the imitator received a license from the innovator, but if this product is included, the results do not change in any significant way. Interaction terms were also included in equation (2), but were not significant.

Each of the regression coefficients in equation (2) has the expected sign, is statistically significant, and is large enough so that the effect of each independent variable is substantial. If R_i increases by 20 percentage points, C_i decreases by about 0.14. If the i th product is a new drug, C_i is about 0.31 larger than would otherwise be the case. If the i th product is a new use for an existing material on which a firm other than the innovator holds the patents, C_i is about 0.54 smaller than would otherwise be the case. These three independent variables can explain about half of the variation in C_i .

IV. PATENTS AND IMITATION COSTS

Still another factor that affects the ratio of imitation cost to innovation cost is whether or not the innovator has patents on the new product. Contrary to popular opinion, patent protection does not make entry impossible, or even unlikely. Within 4 years of their introduction, 60 % of the patented successful innovations in our sample were imitated. Nonetheless, patent protection generally increases imitation costs. To obtain information concerning the size of this increase, the firms in our sample were asked to estimate how much the value of C_i for each patented product increased because it was patented. The median estimated increase in C_i was 11 %. They also were asked to estimate how much the value of C_i for each unpatented product would have increased if it had been patented. The median estimated increase in C_i was only about 6 %. (Indeed, for 2 of these products, patent protection would have reduced the money and time required for imitation because in these cases the innovator could keep secret the essential information underlying the product, whereas if it patented it, some of the information would have been disclosed.) The fact that a patent resulted in a larger increase in the imitation costs of the patented products than of the unpatented products was, of course, a major reason why some products were patented and others were not.¹

In the ethical drug industry, patents had a bigger impact on imitation costs than in the other industries, which helps to account for survey results² indicating that patents are regarded as more important in ethical drugs than elsewhere. The median estimated increase in C_i due to patent protection was about 30 % in ethical drugs, in contrast to about 10 % in chemicals and about 7 % in electronics and machinery. Without patent protection, it frequently would have been relatively cheap (and quick) for an imitator to determine the composition of a new drug and to begin producing it. However, for many of these electronics and machinery innovations, it would have been quite difficult for imitators to determine from the new product how it is produced, and patents would not add a great deal to imitation cost (or time).³ These results are in accord with Taylor

¹ Since some of the products in the sample were inherently unpatentable, they had to be omitted, so the results reported in this paragraph are based on 43 products. Using the Wilcoxon test, the difference between the median increase in C_i for patented and unpatented products is significant at the 0.05 probability level. In the entire sample of 48 products, the percentage that was patented was 81 (drugs), 63 (chemicals), and 69 (electronics and machinery).

² For a description of these surveys and their results, see Taylor and Silbertson (1973) and Scherer (1977).

³ Taylor and Silbertson (1973) point out that a high proportion of patents in electronics are thought to be of doubtful validity, and that, even if an electronics patent is valid, it may afford little

and Silbertson's (1973) conclusion that the lack of patent protection would reduce the rate of expenditure on innovative activity to a greater extent in drugs than in other industries.

V. IMITATION COSTS, ENTRY, AND CONCENTRATION

We turn now from the determinants of imitation costs to their effects on entry and concentration. Holding constant the discounted profit (gross of the imitation cost) that the imitator expects to earn by imitating a new product, the new product is more likely to be imitated if the imitation cost is small. To discourage entry, the innovator may adopt pricing (and other) policies to reduce the imitator's expected discounted gross profit if the imitation cost is low. Taking this into account, is it still true that the probability of entry is inversely related to the size of the imitation cost? To find out, we determined whether each product in the sample was imitated within 4 years¹ after it was first introduced. (Innovations that had been on the market less than 4 years and unsuccessful innovations clearly had to be omitted.) Then we carried out a logit analysis to determine whether C – the ratio of imitation cost to innovation cost – influences the probability that entry of this sort occurred within 4 years. Letting P be the probability that such entry did *not* occur, we found that

$$\ln \left(\frac{P}{1-P} \right) = -3.10 + 3.92C. \quad (3)$$

(2.04) (1.97)

Thus, imitation cost seems to be related in the expected way to whether or not entry occurs.²

Imitation cost may also affect an industry's level of concentration. We would expect an industry's concentration level to be relatively low if its members' products and processes can be imitated easily and cheaply.³ For each of the 16 detailed industries included in the sample, we calculated the mean value of C_i , the mean for the j th industry being denoted by $\bar{C}_j$. Then to estimate the relationship between the mean imitation cost and the concentration level, we

protection. See Taylor and Silberston (1973), chapter 12. For some ways (other than patents) in which electronics firms protect their products from imitation, see *Science*, 202, November 24, 1978, pp. 848–9. Also see Scherer (1977).

Although the median increase in imitation cost due to patents (outside the drug industry) is rather small, the increase in some cases is very substantial. Outside drugs, patents increased imitation cost by 100 % or more in about one-quarter of the cases. This helps to explain our findings in Section VI below.

¹ In the bulk of the cases, the new product could have been imitated in 2 years or less even if the imitator carried out the project at the most leisurely pace. In practically all cases it could be imitated in 3 years or less. Thus, 4 years was plenty of time for an imitator to enter.

² Note that a one-tailed test is appropriate here. The methods used to estimate this equation are described in Berkson (1953). For an analysis of entry rates in the chemical industry that estimates the effects of certain kinds of imitation costs, see Mansfield *et al.* (1977*b*), pp. 119–22. For some evidence that imitation costs influence the gap between social and private rates of return from innovations, see Mansfield *et al.* (1977*a*).

³ In their simulation model, Nelson and Winter (1978) use a hypothetical 'ease of imitation factor' which is defined differently from our C_i , but it seems clear that this factor is closely related to what our C_i measures.

regressed each industry's 4-firm concentration ratio, K_j , on $\bar{C}_j$, the result being¹

$$K_j = 6.22 + 61.5\bar{C}_j. \quad (4)$$

(8.80) (12.8)

This finding, which seems to be the first empirical evidence regarding the relationship between the ease of imitation and the level of concentration, is entirely consistent with this hypothesis. Given the large number of factors influencing an industry's concentration level, it is interesting that this relationship is relatively close ($r^2 = 0.60$). Differences among industries in the technology transfer process (including transfers that are both voluntary and involuntary from the point of view of the innovator) may be able to explain much more of the interindustry variation in concentration levels than is generally recognised.

VI. PATENTS AND THE RATE OF INNOVATION

Finally, we turn to one of the most important and controversial questions concerning the patent system: what proportion of innovations would be delayed or not introduced at all if they could not be patented? To shed light on this question, we asked each innovating firm whether it would have introduced each of its patented innovations in our sample if patent protection had not been available. Although answers to such questions have obvious limitations and must be treated with caution, they should shed some light on this topic, about which so little is known.² According to the firms, about one-half of the patented innovations in our sample would not have been introduced without patent protection. The bulk of these innovations occurred in the drug industry. Excluding drug innovations, the lack of patent protection would have affected less than one-fourth of the patented innovations in our sample.

Taylor and Silberston (1973) have provided estimates of the proportion of R and D expenditures in various U.K. industries that would not have been carried out without patent protection. Applying the industry weights in our sample to their data, one would expect that about 36 % of the R and D expenditures of the firms in our sample would not have been carried out under these circumstances, if U.S. and British firms are alike in this regard. This is somewhat less³ than the proportion of innovations that, according to the firms, would not have been introduced without patent protection.⁴

¹ To prevent confusion, note that each concentration ratio refers to all the detailed industry's products, not just the new products in our sample. In effect, we treat the products in our sample in each detailed industry as a sample of all the detailed industry's products, and use $\bar{C}_j$ as a measure of the ease with which all the latter products can be imitated.

² These data could be obtained for 31 innovations. Some innovations could not be included because they were unpatentable or because no reliable information could be obtained from the firm. For a good discussion of what currently is known about this topic, see Scherer (1977 and 1980) and Taylor and Silberston (1973).

³ This difference is statistically significant at the 0.05 probability level.

⁴ There is no reason to expect the proportion of R and D expenditures that would not be carried out without patent protection to equal the proportion of innovations that would not be carried out under these circumstances. If the innovations that would not be carried out tend to be less R and D-intensive than those that would be carried out under these circumstances, the former proportion would be less

One important reason why patents frequently are not regarded as crucial is that they often have only a limited effect on the rate of entry. For about half of the innovations, the firms felt that patents had delayed the entry of imitators by less than a few months.¹ Although patents generally increased the imitation costs, they did not increase the costs enough in these cases to have an appreciable effect on the rate of entry. But although patent protection seems to have only a limited effect on entry in about half of the cases, it seems to have a very important effect in a minority of them. For about 15 % of the innovations, patent protection was estimated to have delayed the time when the first imitator entered the market by 4 years or more.

According to many economists, patent protection tends to be more important to smaller firms than to larger ones. However, although this proposition seems reasonable, the existing evidence on this score is weak and sometimes contradictory.² To test this proposition, we carried out a logit analysis to see whether π , the probability that an innovation would have been introduced without patent protection, is related to S , the 1976 sales of the firm that carried out the innovation. Since practically none of the drug innovations would have been introduced without patent protection, such innovations were excluded.³ The results are:

$$\ln \left(\frac{\pi}{1-\pi} \right) = 0.878 + 0.00012S. \quad (5)$$

(1.03) (0.030)

Since the coefficient of S is far from statistically significant, there is no evidence that patent protection was more likely to be deemed essential for innovations carried out by smaller firms than for those carried out by larger ones.⁴

VII. CONCLUSION

At a purely theoretical level, imitation costs have long played a major role in economic models of innovation and technological change, but systematic empirical investigations have been lacking. While the present study is only a relatively small-scale undertaking with obvious limitations, we believe that it

than the latter. Also, there is no reason to expect that American and British firms would behave in exactly the same way. Thus, if patents are regarded as more important in the United States than in the United Kingdom, this would result in the sort of difference we observe. In addition, there are sampling errors in both our estimates and those of Taylor and Silberston.

¹ Each innovating firm was asked to estimate how much sooner the first imitating firm would have entered the market if the new product had not had patent protection. The resulting estimates are rough, but, according to the firms, they are reasonably accurate. They seem consistent with the findings in the first paragraph of this section. Outside drugs, patents were estimated to delay the entry of imitators by 2 years or more in 25 % of the cases.

² See Scherer (1977).

³ Had the drug innovations been included, a bias might have resulted, since, as noted above, patents are regarded as particularly important in the drug industry. Thus, since the drug firms were larger than the average in the sample, this might have resulted in a spurious inverse relationship between π and S . S is measured in billions of dollars.

⁴ These results may be due in part to the fact that practically all of the innovating firms in the sample are quite large. If a larger number of very small firms were included, the results might be more in line with the proposition in the text.

demonstrates both that empirical investigations of this topic can be carried out and that, unless such investigations are carried out, we are unlikely to understand many fundamental aspects of the economics of technological change. In our view, there is a considerable need for more studies of the size, determinants, and effects of imitation costs.

Our findings should help to promote the search for more realistic and useful models of the innovative process. In recent years, there has been a tendency for such models to assume that the innovator receives all of the benefits from an innovation and that imitation can be ignored. (For example, see Dasgupta and Stiglitz (1980) and Loury (1979).) And in studies of optimal patent life, it is often assumed that the patent holder is free from imitation for the life of the patent. Although we understand how convenient such assumptions may be, our results suggest how considerably they depart from reality. (Recall, for example, that about 60 % of the patented innovations in our sample were imitated within 4 years.) We hope that the excellent theorists who are working in this area will soon be able to relax these assumptions.¹

This study also sheds new light on the effects of the patent system. Contrary to the assumption of many economic models, a patent frequently does not result in a 17-year monopoly over the relevant innovation. Patents do tend to increase imitation costs, particularly in the drug industry, but excluding drugs, patent protection did not seem essential for the development and introduction of at least three-fourths of the patented innovations studied here. From the point of view of public policy, this obviously is an interesting finding.

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¹ The innovator may receive far less than other firms that subsequently introduce the innovation in a somewhat different way or at a more propitious time. Even if the innovator is very successful, imitators may gain a substantial share of the benefits. Models that ignore considerations of this sort should be viewed with caution, since (among other reasons) they omit factors that are important impediments to firms' appropriation of the benefits from their innovations. By doing so, they may exaggerate the incentive for firms to invest in R and D and other innovative activities.

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